Executive Summary

This report investigates how aircraft heavy-maintenance schedules can be improved to protect fleet availability over a long planning horizon. When several aircraft enter maintenance at the same time, the number available for operations in the fleet will fall. The aim of this project was to reduce these bottlenecks by moving maintenance checks within their legal start windows, while preserving the required operational rules for each aircraft.

The work began with a simple neighbourhood search on a small fleet dataset. This established that maintenance start dates could be shifted to improve the availability profile while remaining inside the required maintenance windows. The model was then expanded to a heavier forecast containing 24 aircraft and 278 maintenance checks. This larger dataset included A, B, C and D checks with different durations, intervals and early-start allowances, creating a more realistic scheduling problem.

Several search approaches were developed and compared. A full neighbourhood search tested every available move but became slow as the forecast grew. Random sampling reduced runtime by evaluating a smaller set of possible moves, although it could miss useful schedule changes. The final priority-based approach concentrated on checks that overlapped the lowest-availability days. Rather than searching every possible move equally, it focused effort on the maintenance checks contributing directly to the main operational problem.

The priority-based approach produced the strongest practical result. It increased worst-day availability from 15 to 18 aircraft and reduced the variation in daily availability from approximately 1.95 to 1.41. It also reduced the runtime from approximately 15 minutes for full neighbourhood search to around 15 seconds for the original priority-based result. These improvements were achieved by rebuilding later maintenance checks whenever an earlier check moved, ensuring that the resulting schedule remained consistent with the maintenance rules.

The project then investigated whether simulated annealing could improve the priority-based result further. Simulated annealing allows a search to occasionally accept a worse temporary solution, which may help it escape a locally good but ultimately limited schedule. The approach produced a very small improvement in schedule smoothness but did not improve worst-day availability beyond 18 and required substantially longer runtime. The tuned priority-based method was therefore retained as the preferred approach for the tested forecast.

A further feasibility analysis showed that availability of 19 aircraft cannot be achieved every day under the current maintenance rules and forecast. Six long D checks have unavoidable overlap during May 2033, meaning that at least six of the 24 aircraft must be in maintenance

on those days. The maximum possible availability is therefore 18 aircraft. Since the priority-based method reached this value, it achieved the best possible worst-day availability for the current scheduling instance.

After reaching this availability limit, the project shifted focus from further algorithm changes to the way the fleet was initially staggered. The original heavy-maintenance forecast contained a concentration of long D checks in a similar part of the planning horizon. Since these D checks last 126 days, their overlap created the major availability bottleneck that limited the original forecast to 18 available aircraft. To investigate whether this problem could be reduced earlier in the planning process, two alternative starting schedules were created: a moderate stagger and a wider stagger. These schedules spread the initial maintenance ages of aircraft more evenly, causing D checks to occur across a broader range of dates rather than becoming concentrated in one period.

The results showed that the initial stagger had a major effect on the quality of the final schedule. The original as-loaded schedule reached a final worst-day availability of 18 aircraft with a standard deviation of approximately 1.414. The moderate stagger reached 19 aircraft with a standard deviation of 0.944, while the wide stagger reached 20 aircraft with a standard deviation of 0.713. This indicates that the starting maintenance distribution can have a greater influence on long-term fleet availability than further changes to the search algorithm. Although the optimiser can improve an existing schedule, it cannot fully remove bottlenecks created by tightly grouped D checks.

A hyperparameter analysis was then performed to identify whether different neighbourhood step sizes could further improve the priority-based approach. The analysis compared multiple combinations of maintenance-date shifts across the original, moderate and wide forecasts. The strongest configuration used step sizes of 56, 42, 28, 14, 7, 3 and 1 days. This configuration improved schedule smoothness on all three forecasts while retaining the same worst-day availability. For the original forecast, the standard deviation reduced from 1.4221 to 1.4019. The moderate stagger improved from 0.9117 to 0.8834, while the wide stagger improved from 0.6167 to 0.5868. These results show that careful tuning can improve the smoothness of the maintenance schedule, although the initial stagger remains the most important factor in preventing major availability bottlenecks.

Overall, the project demonstrates that targeted local search using metaheuristic algorithms can improve fleet maintenance flow quickly and effectively. The developed method provides a practical foundation for future work involving additional operational constraints, live planning data and disruption scenarios such as unscheduled maintenance.

Conclusion

This project set out to investigate whether aircraft maintenance checks could be scheduled more effectively to reduce bottlenecks and improve fleet availability. The work showed that the main difficulty is not simply finding an improved start date for one check. Each change can affect later maintenance requirements, and the final schedule must still remain inside the legal maintenance windows for every aircraft. The project therefore required a method that could improve availability while preserving the full maintenance sequence.

The final priority-based selection method achieved this purpose. By concentrating on the checks that affected the lowest-availability days, it avoided the excessive computation required by full neighbourhood search and avoided the uncertainty introduced by random sampling. The method provided a practical way to improve the schedule while maintaining the required maintenance rules. Simulated annealing produced only a minor improvement in smoothness at a much higher runtime, so priority-based selection remained the preferred method for the tested forecast.

The project also showed that some availability problems cannot be solved through optimisation alone. The original forecast contained overlapping D checks that created an unavoidable limit of 18 available aircraft during May 2033. This result is important because it distinguishes a limitation in the search method from a limitation in the forecast itself. Repeating the search for longer or using a different algorithm would not remove maintenance overlap that is required by the existing check durations and start windows.

The initial-stagger experiments provided a broader lesson. When aircraft entered the forecast with more evenly distributed maintenance ages, the optimiser achieved stronger final availability and smoother schedules. The moderate and wide staggers reached worst-day availability of 19 and 20 aircraft respectively. This suggests that long-term maintenance planning should begin before D checks become concentrated in the same period. Optimisation can improve an existing schedule, but an early maintenance-staggering strategy can prevent some bottlenecks from developing in the first place.

The hyperparameter analysis further supported the final method by showing that the choice of neighbourhood step sizes affects how effectively the optimiser improves a schedule. Larger moves allowed the search to make meaningful changes early, while smaller moves refined the final result. The selected combination of 56, 42, 28, 14, 7, 3 and 1-day shifts improved smoothness across the tested forecasts without reducing worst-day availability.

Overall, the project developed a practical foundation for improving fleet maintenance scheduling. It showed that priority-based selection can quickly identify and reduce availability bottlenecks, while the initial stagger results demonstrated the value of planning maintenance flow over a longer horizon. Further work should test the method using operational data and include factors such as maintenance capacity, workforce availability, component delays and unscheduled maintenance.